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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Load the image
img = cv2.imread('image1.jpg')

if img is not None:
    # Laplacian mask with negative center coefficient
    # This kernel detects edges/details: [[0, 1, 0], [1, -4, 1], [0, 1, 0]]
    # To sharpen, we subtract the Laplacian from the original: Original - Laplacian
    # Which is equivalent to applying a kernel like: [[0, -1, 0], [-1, 5, -1], [0, -1, 0]]

    laplacian_kernel = np.array([[0, 1, 0],
                                  [1, -4, 1],
                                  [0, 1, 0]], dtype=np.float32)

    # Calculate the details (Laplacian)
    details = cv2.filter2D(img, -1, laplacian_kernel)

    # Sharpen by subtracting the Laplacian (negative center mask) from the original
    # Since the center is -4, subtracting it effectively boosts the center
    sharpened = cv2.subtract(img, details)

    # Save and display results
    cv2.imwrite('laplacian_sharpened.jpg', sharpened)
    print("Original Image:")
    cv2_imshow(img)
    print("\nSharpened Image (Laplacian with Negative Center Mask):")
    cv2_imshow(sharpened)
else:
    print("Error: 'image1.jpg' not found.")
```

Original Image:



Sharpened Image (Laplacian with Negative Center Mask):

